Building a Predictive Model for Credit Risk Evaluation with Machine Learning

IT745-A Project

By

Timothy T Ornguze

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I. Abstract

Over the Years, financial institutions have relied on various methods to determine if customers can make payments on their loans. These methods were ineffective at best, and some were not rooted in science, which increases their risk exposure due to non-payment of loans. According to Danenas et al. (2010), the loan is a critical topic in the banking and financial industry because the process of issuing the loan requires much attention to ensure the possibility of getting the funds back

We would employ a Machine learning approach to simplify and enhance the accuracy of the process. We will utilize an individual's credit history and income levels, then fit these data into a linear regression model to obtain a risk score. The result will enable us to predict whether an individual will default on their loan. We would use the credit Risk data to train our AI model, dividing the data into two. The training set is used to teach the model about the relationships between the input and target variables, and the validation set is used to assess how well the model processes new data.

II. Introduction

We would build a machine learning model to predict loan status using a credit risk dataset. This Model can be used to determine the risk level of customers in a financial institution or anyone who seeks to obtain a loan. The model can be used to predict whether the borrower will default on the loan or not.

Gui (2019) states that financial institutions should have a risk prediction model to enable them to predict the characteristics of individuals who default on loans. We would create a Machine learning model to determine the credit risk of an individual using readily available data. We use the machine model to load the credit risk dataset, clean the data, encode the categorical variables, and build the machine learning model using RandomForestClassifier. GridSearchCV will be used to tune hyperparameters and improve model performance.

III.Methodology

We built a machine learning model. First, we import libraries for our data processing using pandas, NumPy, and Matplotlib. pyplot for visualization, as well as model training (RandomForestClassifier), evaluation (classification_report and confusion_matrix), and Hyperparameter tuning (GridSearchCV). Upload the credit risk dataset into our machine learning model. We performed an initial data clean-up, writing out a formula to refine the data. We then proceeded to encode the categorical variables and conducted a final data clean-up. We trained and split our data into X and Y variables, splitting the data into a testing set (75%) and a training set (25%).

Train a baseline forest model, RandomForestClassifier. We print output and obtain output accuracy, precision, recall, F1 score, and a confusion matrix. We tune the model using GridSearchCV and perform a 5-fold cross-validation to find the best model configuration. We then print the tuned models with improved performance and select the best hyperparameters. Lastly, we plot a graph using the tuned data.

1V. Results

Class zero has a high precision and recall value (0.92, 0.99)

Class 1 also has a high precision value (0.95) but a lower recall (0.71). This may be due to the model missing a small number of actual defaults.

For the confusion matrix: Baseline Model

	precision	recall	f1-score	support
0	0.92	0.99	0.95	5701
1	0.95	0.71	0.82	1666
accuracy			0.93	7367
macro avg	0.94	0.85	0.89	7367
weighted avg	0.93	0.93	0.92	7367

[[5643 58]

[479 1187]]

5643 non-defaults were correctly predicted.

58 non-defaults wrongly predicted as defaults (false positives).

479 defaults missed (false negative).

1187 defaults were correctly predicted.

Tuned Model performance after GridSearchCV

	precision	recall	f1-score	support
0	0.91	0.99	0.95	5701
1	0.96	0.68	0.79	1666
accuracy			0.92	7367
macro avg	0.93	0.84	0.87	7367
weighted avg	0.92	0.92	0.92	7367

[[5648 53]

[533 1133]]

After the model was tuned the accuracy has slightly improved, still 92-93%.

There was a slight drop in the minority class from 0.71 to = 0.68.

False negatives had a slight increase from 479 to 533

Confusion matrix:

5648 correct non-defaults

53 false positives (slightly better)

533 false negative (slightly worse)

1133 react defaults

V. Conclusion

The Baseline model performed well. The tuned model did not do significantly better than the baseline model in terms of recall for class 1. Precision improved slightly, and there was a reduction in false positives.

The 10 most important features from the machine learning model that was used to plot the graph are: Loan_percent_income, Loan_grade, person_home_ownership, person_income, loan_int_rate, loan_amnt, person_emp_length, cb_person_default_on_file, and person_age.

From the model, we can infer that loan_pervent_income is the most significant factor in determining loan default, while person_age is the least important. Financial institutions may utilize this model to mitigate their risk exposure associated with loan default. This project seeks to improve the accuracy of credit risk using available data to build a machine learning model.

More work could be done to address the class imbalance, making use of class_ weight =’ imbalanced' in RandomForestClassifier, and oversampling.

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